Candy Lottery

Time limit: 1.00 s
Memory limit: 512 MB

CSES 1727 | Mathematics | https://cses.fi/problemset/task/1727

There are n children, and each of them independently gets a random integer number of candies between 1 and k.

What is the expected maximum number of candies a child gets?

Input

The only input line contains two integers n and k.

Output

Print the expected number rounded to six decimal places (rounding half to even).

Constraints

- $1 \le n \le 100$
- $1 \le k \le 100$

Example

Input
2 3

Output
2.444444